

Lesson Exemplar in General Science

Quarter 3

LE

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Lesson Exemplar for General Science
Quarter 3: Unit 2

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LESSON EXEMPLAR

Learning Area	General Science	Grade Level	11
Semester	2 nd	Quarter	3

I. OBJECTIVES (*Identifying the Goals*)

Content Standard	The learners learn that cells carry out functions that enable organisms to survive.
Performance Standard	By the end of the quarter, learners use examples to describe the relationships among living systems and their components and how the structure of a part relates to its function. They explain the role of cells in perpetuating life and how organ systems work together to maintain optimal health. They conduct investigations on the effects of climate change on local ecosystems.
Learning Competencies	Key Topics Learned in the Previous Grade Level: 1. Grade 7 (Life Science): Learners studied the characteristics of living things and were introduced to the concept that the cell is the basic unit of life. They learned that organisms require nutrients, water, and energy to survive. 2. Grade 8 (Biology): Learners explored the parts of the cell and their basic functions, including the nucleus, cell membrane, and cytoplasm, which regulate and support cellular activities. 3. Grade 9 (Biology): Learners learned about photosynthesis and respiration, emphasizing how plants produce their own food and how both plants and animals release energy from food through respiration. They also touched on the concept of energy flow in ecosystems. These foundational topics from Grades 7–9 serve as scaffolding for this lesson. In this lesson, grade 11 learners <i>describe how cells obtain nutrients from food and convert them into energy through cellular processes to sustain life in plants and animals</i>. At this stage, learners deepen their understanding by focusing on how cells obtain nutrients and convert them into usable energy (ATP) through cellular processes such as photosynthesis and cellular respiration. Teachers should connect these prior learnings to highlight continuity and avoid redundancy.



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II. REFERENCES and MATERIALS <i>(Selecting Resources and Material)</i>	Department of Education. (2024). Senior High School Curriculum Guide: General Science (MATATAG Curriculum). DepEd Central Office.	
	Campbell, N. A., Urry, L. A., Cain, M. L., Wasserman, S. A., Minorsky, P. V., & Reece, J. B. (2021). Biology: A Global Approach (12th ed.). Pearson.	
	Alberts, B., Johnson, A., Lewis, J., Raff, M., Roberts, K., & Walter, P. (2022). Molecular Biology of the Cell (7th ed.). Garland Science.	
	Open Educational Resources (OER) – YouTube: Amoeba Sisters. (2017). Photosynthesis and Cellular Respiration. https://youtu.be/2f7YwCtHcgk	
	Philippine Science Teacher Journal (Recent issues on biology teaching strategies).	
	Instructional materials: PPT, laboratory/activity materials, printed worksheet	
<i>(These shall be accomplished per topic)</i>		
III. CONTENT <i>(Sequencing Content)</i>	The Importance of Cells in Living Things	
IV. OBJECTIVES <i>(Setting Clear Objectives and Analyzing the Tasks)</i>	At the end of the lesson, the learners will be able to:	
	1. Define the cellular processes that enable cells to absorb nutrients and produce energy necessary for sustaining life in plants and animals. 2. Explain how nutrients from food are absorbed and transported to cells in plants and animals, using real-life examples. 3. Describe the role of cellular respiration in converting nutrients into energy (ATP), and relate this to everyday biological functions like movement and growth..	
IV. PROCEDURES <i>(Selecting Strategies, Making Meaningful Content, Delivering Lesson and Assessing Learning)</i> This section focuses on selecting learner-centered, evidence-based instructional approaches such as problem-based learning, collaborative tasks, interdisciplinary integration, and		ANNOTATION <i>*Instruction to teacher on how to facilitate the activities.</i>



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technology-enhanced instruction. These strategies are intended to foster active engagement, critical thinking, and adaptability across diverse learning pathways. The chosen approaches and methodologies will be reflected through varied and relevant activities and assessments that emphasize real-world relevance and application, thereby enhancing learner engagement and comprehension. (Each part shall have 2-3 varied activities)	<i>*In the Annotation, explicitly explain how the IDF is applied in each part of the lesson</i>
A. Activating Prior Knowledge 1. Activating Prior Knowledge Activity 1: Cell Sprint (Interactive Quiz Game) Objective: To recall prior knowledge on cell parts and functions, photosynthesis, and respiration through a fun ICT-based quiz. Learners join a Quizizz game with 15 varied-difficulty questions. The leaderboard motivates students and highlights recall gaps. Suggested link for the quiz: https://tinyurl.com/2nk8bhdw Activity 2: Think-Pair-Share Objective: To allow learners to connect their everyday experiences of food/energy to cellular processes. Teacher asks: 'How do you think plants and humans get energy from the food they take in?' Students think individually, pair with a seatmate to discuss, and share with the class. If the students	<i>Each part has 2-3 OPTIONS of activities, the teacher may select 1, 2, 3 or may make his own. If you choose any of the activity, make sure to process and connect this to next activity to ensure continuity of learning. In Cell Sprint activity, teacher sets up the quiz and encourages participation. IDF Application: This activity activates learners' schema by linking previous knowledge with new learning. It maximizes engagement (responsive instruction) and creates a cognitive bridge for deeper understanding. The teacher may also make his/her own quiz.</i> <i>Teacher facilitates sharing and validates correct connections. IDF Application: Encourages learner interaction and collaborative meaning-making (relevant instruction). Builds confidence by valuing prior ideas.</i>



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struggled to answer the main question, rephrase it:
Guide Questions if learners struggle:

1. What happens to the food you eat once it enters your stomach?
2. Where do you think the energy from rice, bread, or fruits goes in your body?
3. How do plants get their 'food' if they don't eat like animals?
4. What do you think happens to the sunlight that plants absorb through their leaves?
5. Have you experienced feeling weak when hungry? What do you think your cells are missing at that time?

Activity 3: KWL Chart

Objective: To assess what learners already know and want to know about cellular energy processes.

Students fill in the 'K' (Know) and 'W' (Want to know) parts of a KWL chart. The 'L' (Learned) part is to be completed at the end of the lesson.

Instruction: Fill in the columns below before, during, and after the lesson.

K (What I Know)	W (What I Want to Know)	L (What I Learned)

Teacher distributes chart templates and explains its purpose. IDF Application: This scaffolds metacognition (reflective instruction), ensuring learners track knowledge before and after instruction.



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2. Establishing the Purpose of the Lesson

*Prior to activity/ies, make sure to introduce the learning objectives. You may opt to let the students copy the objective or distribute printed copy of it in Likert Rating Scale format, so they can assess their own level of achieving the lesson objectives.

Activity 1: Motivational Question

Objective: To spark curiosity and connect the lesson to learners' real-life experiences by exploring the role of food and energy in daily activities.

Teacher asks: 'Why do you feel weak when you skip meals?'

If students struggle to respond, the teacher can guide them with follow-up prompts:

1. "What happens to your body when you haven't eaten breakfast?"
2. "How does eating rice or bread help you play basketball or concentrate in class?"
3. "Why do athletes or workers eat before doing hard activities?"

After student responses, the teacher links them to the idea that cells obtain nutrients from food and convert them into energy, setting up the purpose of the lesson.

The teacher should validate all student answers and connect them back to cellular processes.

IDF Application: This activity establishes relevance (Relevant Instruction) by showing why the topic matters in everyday life. It answers the question "Why are we learning this?" and motivates learners to engage in the upcoming activities.



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	Activity 2: Framing the Goal Teacher presents the day's objectives in student-friendly terms. Learners write a personal goal (e.g., 'I want to understand how my cells get energy from food').	<i>Teacher emphasizes alignment of class goals with student goals. IDF Application: Ensures constructive alignment by clarifying objectives, promoting learner ownership of learning.</i>
B. Instituting New Knowledge	1. Presenting Examples Activity 1: Food Journey Story Map Objective: Learners trace how nutrients from food travel through the body to reach the cells The teacher presents a “food journey” scenario. Learners create a flow chart showing how food is digested, absorbed, transported, and used by cells. <i>Sample Scenario:</i> “Miguel, a Grade 11 basketball player, has a big championship game in the afternoon. At lunchtime, he eats a plate of rice, fried chicken, and a glass of orange juice. Later, during the game, he feels energized and able to run quickly across the court.” Use this guide for if the student will have a hard time making the flow chart. Step 1: Digestion What happens to the rice, chicken, and orange juice when they are broken down into simpler nutrients? Step 2: Absorption Where in the body are these nutrients absorbed into the bloodstream?	<i>Teacher facilitates brainstorming, provides chart paper, and checks flowchart accuracy. IDF Application: Provides relatable, real-life examples (Presenting Examples) to contextualize abstract processes and engage learners</i>



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	Step 3: Transport How are the nutrients transported to the muscle cells? Step 4: Conversion to Energy What happens inside the cell (mitochondria) when nutrients are converted into energy Activity 2: Plant Energy Pathway Demo Objective: Learners visualize how plants acquire and transform nutrients Teacher demonstrates using cut-outs or drawings: a leaf, arrows for sunlight, CO₂, and H₂O. Learners explain how these inputs produce glucose and oxygen. Activity 3: Video Case Analysis Objective: Learners identify the complementary relationship between photosynthesis and respiration Teacher shows short video clip (Photosynthesis and Respiration, YouTube, ~5 minutes). Learners fill in a two-column table: Inputs/Outputs of Photosynthesis vs. Respiration. Groups present findings. Suggested video: https://www.youtube.com/watch?v=8B_64G2SI_8 Title: Photosynthesis and Respiration Author/Channel: Uploaded on YouTube Teacher's Pet	<i>Teacher prompts with 'What comes in? What goes out?' IDF Application: Makes abstract ideas concrete by using visual prompts, supporting diverse learners.</i> <i>Teacher pauses video at key points, asks clarifying questions. IDF Application: Uses multimedia as example (Responsive Instruction) to make processes visible and relatable.</i>
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	Duration: 1.27 minutes While watching, learners are asked to take note of: • The inputs and outputs of photosynthesis • The inputs and outputs of cellular respiration • The connection between the two processes After viewing, the teacher facilitates discussion with guide questions: 1. What do you notice about the relationship between photosynthesis and respiration? 2. How do these two processes complement each other in the survival of living things? 3. Why do you think plants and animals are interdependent when it comes to energy? 2. Discussing New Concept Activity 1: Mini-Lecture with Diagramming Objective: Learners explain how cells convert glucose into energy (ATP) through cellular respiration Teacher provides a simplified diagram of glycolysis, Krebs cycle, and ETC. Learners label and annotate the diagram. Activity 2: Group Concept Mapping Objective: Learners show how organelles (chloroplasts, mitochondria) support energy conversion : In groups, students create a concept map linking organelles to processes. Example: 'Chloroplast → Photosynthesis → Glucose' and 'Mitochondria → Respiration → ATP'.	<i>Teacher explains in simple terms, connects each step to energy production. IDF Application: Encourages guided exploration of complex processes.</i> <i>Teacher provides sample linking words and supports weaker groups. IDF Application: Promotes collaborative meaning-making and visualization of relationships.</i>
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	Activity 3: Q&A Chain Objective: Learners clarify misconceptions by answering peer-generated questions. Each learner writes one question about energy processes, exchanges with a peer, and answers. Teacher clarifies misconceptions in plenary. 3. Developing Mastery Activity 1: Diagram Construction Objective: Learners construct diagrams of nutrient pathways in plants vs. animals Teacher provides blank diagram templates for plants and animals. Learners fill in processes: nutrient absorption, photosynthesis, respiration, ATP production. Activity 2: Group Debate: 'Which System is More Efficient?' Objective: Learners critically analyze differences in nutrient-energy acquisition Class is divided into two groups – Team Plant and Team Animal. Each defends efficiency of their system using evidence from lessons. Teacher moderates debate. Debate Type: Modified Oxford-Style Debate	<i>Teacher circulates, listens to exchanges, and addresses misconceptions. IDF Application: Builds reflective instruction by promoting learner inquiry and peer teaching.</i> <i>Teacher checks accuracy of outputs/inputs. IDF Application: Learners independently apply knowledge in visual form, reinforcing mastery.</i> <i>The teacher acts as moderator, ensuring respectful turn-taking and equal speaking time. The teacher uses the guide questions to stimulate deeper analysis when arguments stall.</i> <i>IDF Application: This debate promotes critical thinking (Developing Mastery) and encourages learners to apply knowledge in a collaborative and evaluative context. It connects classroom</i>
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	2 Teams: Team Plant (arguing plants are more efficient because they produce their own food) and Team Animal (arguing animals are more efficient in nutrient/energy use). Time Allocation: • 3 minutes Opening Statement per team • 2 minutes Rebuttal per team • 2 minutes Open Forum (moderated Q&A) • 2 minutes Closing Argument per team Scenario: <i>“Imagine a world facing food scarcity. Which living system—plants or animals—has a better strategy for sustaining energy and survival? Debate your stance using scientific evidence.”</i> Moderator’s Guide Questions: (To ensure the debate stays on track and aligned to the objective) • For Team Plant: “How do photosynthesis and the ability to produce food make plants more efficient in survival?” • For Team Animal: “How does the digestive system and mobility of animals make them more efficient in energy acquisition?” • “What are the weaknesses of your system compared to the other?” • “How does your system adapt to environmental changes such as lack of sunlight or food scarcity?” • In terms of sustaining life on Earth, which system has a greater impact and why?”	<i>learning to real-world concerns like survival and sustainability.</i>
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	Activity 3: Practice Worksheet (What if Scenarios) Objective: Learners analyze effects of disruptions in cellular processes The teacher will give students worksheet with scenarios, e.g., 'What happens if mitochondria stop working in muscle cells?' 'How would a plant survive without sunlight for a week?' Learners answer in pairs.	<i>Teacher checks responses and provides corrective feedback. IDF Application: Reinforces mastery by applying concepts to problem-solving situations</i>
C. Demonstrating Knowledge and Skills	1. Finding Practical Application Activity 1: Hands-On Experiment: Starch Test in Leaves / Yeast Respiration Objective: Learners perform an experiment to demonstrate energy transformation in cells. Option A (Plants): Learners test leaves for starch using iodine. Option B (Respiration): Learners set up yeast-sugar solutions in test tubes with balloons to capture CO₂ from respiration. Expected Output: Lab report with procedure, results, and conclusion. Activity 2: Case Study Analysis: Malnutrition and Energy Deficiency Objective: Learners explain how proper nutrition supports efficient cellular processes (Objective link: Explain importance of nutrition).	<i>Teacher ensures safety and facilitates experiment. IDF Application: Provides real-world, authentic application of science concepts.</i> <i>The teacher facilitates by asking probing questions and ensuring learners link real-life nutrition issues to cellular energy processes.</i>



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Sample Case Study: *Ana, a Grade 11 Academic student, often skips breakfast because she wakes up late. During Life Skills class, she feels dizzy and weak after running only two laps. Her teacher notices she looks pale and advises her to visit the clinic. The school nurse finds that Ana's blood sugar is low, and she also shows signs of iron deficiency.*

Learners are asked to analyze:

- What is happening inside Ana's cells when she lacks glucose and iron?
- How does skipping meals affect ATP production and her energy level?
- What are possible long-term effects on her health and academic performance if this habit continues?
- What advice would you give Ana to improve her nutrition and cellular energy production?

Expected Output: Group report or role-play as nutrition consultants.

Activity 3: Ecosystem Interdependence Simulation

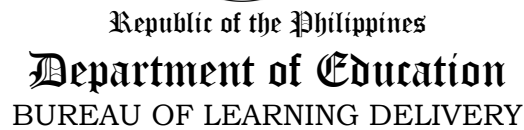
Objective: Learners reflect on the interconnectedness of plants and animals in sustaining energy flow.

What to Do:

1. The teacher prepares role cards labeled: Sun, Plant, Herbivore, Carnivore, Decomposer. (Other learners can join in by holding multiple roles or becoming additional organisms like Omnivores, Humans, etc.).

IDF Application: This case ensures Relevant Instruction by connecting cellular biology to real health experiences of students, making learning both practical and reflective.

The teacher ensures all learners participate, pauses to ask critical thinking questions, and summarizes how energy at the cellular level (ATP production, glucose breakdown, oxygen use) sustains life at the ecosystem level.

[illegible]



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Objective: Learners summarize their main takeaway about energy processes.

Students write on index cards: 'The most important thing I learned today is...'

3. Evaluating Learning

Activity 1: Fact or Bluff

Objective: Learners distinguish correct from incorrect statements about cellular energy processes (photosynthesis, respiration, nutrient acquisition).

Instructions: The teacher reads each statement aloud (or flashes them on a slide). Learners raise a “Fact” or “Bluff” card (or move to sides of the classroom). Teacher immediately reveals the correct answer and explains briefly.

Statements:

RUBRICS

3	2	1	0
Demonstrates clear and accurate understanding of the topic; expresses one big idea that captures the essence of the lesson.	Shows partial understanding ; expresses a relevant but incomplete or somewhat vague idea.	Shows limited understanding ; idea is unclear or contains misconceptions.	No response or response unrelated to the topic.

Teacher prepares clear statements and ensures immediate feedback.

IDF Application: This activity is a quick formative check, engaging all learners actively while reinforcing key knowledge.



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1. The cell is the basic unit of life where energy transformations occur. → Fact
2. Mitochondria are found only in plant cells. → Bluff (they are found in both plant and animal cells)
3. Chloroplasts are the site of photosynthesis. → Fact
4. In cellular respiration, oxygen is required to release energy from glucose. → Fact
5. The main energy currency of the cell is starch. → Bluff (it's ATP)
6. Glucose produced by plants can also be used by animals for energy. → Fact
7. Plants do not need oxygen because they make their own food. → Bluff (plants also perform respiration and need oxygen) Without mitochondria, muscle cells would struggle to produce enough energy for activity. → Fact
8. Photosynthesis and respiration are complementary processes in the energy cycle. → Fact
9. Skipping meals has no effect on cellular energy production. → Bluff (less glucose = less ATP, leading to weakness)

Activity 2: Energy Relay (Team Challenge)

Instructions:

1. The class is divided into 3–4 teams.
2. Each team receives an envelope containing cut-out cards with terms/phrases related to both processes (e.g., Sunlight, CO₂, H₂O, Glucose, Oxygen, ATP, Mitochondria, Chloroplast, Glycolysis, Krebs Cycle, ETC).

The teacher ensures fairness, checks sequencing, and clarifies misconceptions.

IDF Application: This activity promotes collaborative learning (Responsive Instruction)



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	3. On the board/manila paper, two big columns are labeled “Photosynthesis” and “Cellular Respiration.” 4. At the signal, team members take turns running to the board to place one card under the correct process in its proper order. Only one member can place a card per turn (relay style). 5. Teams that finish first and with the most correct sequence win. Expected Output: Correctly sequenced diagrams showing the full input-process-output chain for both photosynthesis and respiration. Activity 3: Four Corners Reflection Teacher assigns each corner of the room a reflective prompt: • The most important thing I learned today • A question I still have • How this lesson connects to real life • One way I can apply this learning Learners move to a corner, share in groups, then 1–2 volunteers report back. 4. Additional Activities Activity 1: Research Extension	<i>and reinforces understanding through active, physical engagement (Relevant Instruction).</i> <i>Teacher ensures each group shares and validates ideas.</i> <i>IDF Application: Encourages reflective instruction, letting students generalize and internalize learning.</i>
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	Learners research how climate change affects photosynthesis and respiration in local ecosystems. Activity 2 – Remediation Task Learners revisit their KWL chart with teacher support to reinforce weak areas.	
V. ASSESSMENT (Assessing Learnings)	• <i>Performance Task: Learners create an Infographic – 'From Food to Energy: How Cells Power Life'. This task will be evaluated using a rubric based on clarity, accuracy, creativity, and relevance.</i> • <i>Summative Test: Give students 15-item multiple choice.</i> (Please see annex 1 for the summative test and performance task.)	
VI. REFLECTION (Feedback and Continuous Improvement)	<i>This section presents the key highlights and challenges encountered by both teachers and learners during the teaching-learning process throughout the unit. It also includes the adjustments made by the teacher to improve instruction.</i>	

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Annex 1. Suggested Assessment

Direction: Read each question carefully and choose your answer from the given options.

1. Miguel, a Grade 11 basketball player, eats rice and bananas before his game. During play, his muscles need more ATP. Which process provides the immediate supply of ATP in his muscle cells?
 - a. Photosynthesis
 - b. Glycolysis
 - c. Osmosis
 - d. Transpiration
2. Ana skips breakfast before school. In her first class, she feels weak and cannot concentrate. Which process in her cells is MOST directly affected?
 - a. Cellular respiration
 - b. Photosynthesis
 - c. Diffusion
 - d. Active transport
3. A farmer leaves his crops in a covered greenhouse with no sunlight for 10 days. The plants begin to wither. Why did the plants weaken?
 - a. They could not perform photosynthesis to produce glucose.



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- b. They could not absorb carbon dioxide.
 - c. They stopped cellular respiration.
 - d. They lacked mitochondria.
4. Students observe bubbles forming on water plants exposed to sunlight in their school pond. What do the bubbles MOST likely contain?
- a. Carbon dioxide
 - b. Oxygen
 - c. Glucose
 - d. ATP
5. A student drinks an energy drink before an exam. The drink contains high levels of glucose. Why does glucose help him?
- a. It directly stores ATP.
 - b. It is broken down in cells to release ATP.
 - c. It makes more chlorophyll.
 - d. It replaces oxygen in respiration.
6. Even at night, plants continue to live without sunlight. Which process allows them to survive during the night?
- a. Photosynthesis
 - b. Respiration in mitochondria
 - c. Transpiration
 - d. Chemosynthesis
7. After sprinting, John breathes heavily. His cells switch to anaerobic respiration when oxygen is low. What by-product builds up in his muscles?
- a. Alcohol
 - b. Lactic acid
 - c. Carbon dioxide
 - d. Oxygen
8. Forests help regulate oxygen and carbon dioxide levels. How does deforestation affect this cycle?
- a. More oxygen will be released.
 - b. Less carbon dioxide will be absorbed by plants.
 - c. More glucose will be produced.
 - d. Respiration in animals will stop.
9. A child with protein deficiency is weak and malnourished. Why does lack of protein affect cellular energy processes?



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- a. Proteins directly provide ATP.
 - b. Proteins build enzymes needed for respiration.
 - c. Proteins replace glucose in mitochondria.
 - d. Proteins store starch in the liver.
10. In a mangrove forest, plants provide oxygen to animals, while animals release carbon dioxide to plants. This is an example of:
- a. Competition
 - b. Interdependence
 - c. Mutation
 - d. Succession
11. Students mix sugar with yeast in a sealed bottle. A balloon attached to the opening inflates. What process caused the inflation?
- a. Photosynthesis
 - b. Anaerobic respiration
 - c. Diffusion
 - d. Osmosis
12. A hiker climbing at high altitude finds it hard to breathe and tires quickly. What cellular process is MOST affected by lack of oxygen?
- a. Photosynthesis
 - b. Aerobic respiration
 - c. Glycolysis
 - d. Diffusion
13. Increased carbon dioxide levels in the atmosphere can benefit plants temporarily. Why?
- a. Plants can use more CO₂ for photosynthesis.
 - b. Plants respire faster in high CO₂.
 - c. Plants stop releasing oxygen.
 - d. Plants grow without glucose.
14. Rice is the staple food in the Philippines. Why does rice provide energy for daily activities?
- a. It contains starch that can be converted into glucose.
 - b. It directly contains ATP.
 - c. It provides oxygen to the body.
 - d. It increases protein production.



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15. Which of the following BEST explains why photosynthesis and respiration are complementary?
- a. Photosynthesis releases CO₂, which respiration uses.
 - b. The products of photosynthesis are used in respiration, and vice versa.
 - c. Respiration produces sunlight for photosynthesis.
 - d. Both occur only in plants.

Answer Key:

Question No.	Correct Answer	Rationalization
1	b. Glycolysis	Glycolysis quickly provides ATP without requiring oxygen, supporting muscle activity during intense exercise.
2	a. Cellular respiration	Without glucose from food, cells cannot undergo respiration efficiently, reducing ATP production.
3	a. They could not perform photosynthesis to produce glucose.	Plants need sunlight to make glucose; without it, they cannot sustain energy needs.
4	b. Oxygen	Bubbles from aquatic plants in sunlight are oxygen released during photosynthesis.
5	b. It is broken down in cells to release ATP.	Glucose is metabolized in respiration to produce ATP, fueling brain and body functions.
6	b. Respiration in mitochondria	At night, plants survive by respiring, using glucose and oxygen to produce ATP.
7	b. Lactic acid	In the absence of oxygen, animal cells undergo anaerobic respiration, producing lactic acid.



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8	b. Less carbon dioxide will be absorbed by plants.	Fewer plants mean less CO ₂ absorption, disrupting the oxygen-CO ₂ balance.
9	b. Proteins build enzymes needed for respiration.	Enzymes that regulate cellular processes are proteins; deficiency impairs metabolism.
10	b. Interdependence	Plants and animals rely on each other for oxygen and carbon dioxide in energy cycles.
11	b. Anaerobic respiration	Yeast ferments sugar anaerobically, producing CO ₂ that inflates the balloon.
12	b. Aerobic respiration	Low oxygen reduces efficiency of aerobic respiration, leading to fatigue at high altitudes.
13	a. Plants can use more CO ₂ for photosynthesis.	Higher CO ₂ allows plants to produce more glucose temporarily, until other limits occur.
14	a. It contains starch that can be converted into glucose.	Starch in rice is broken into glucose, used in respiration to provide energy.
15	b. The products of photosynthesis are used in respiration, and vice versa.	These processes complement each other: photosynthesis provides glucose and O ₂ , respiration provides CO ₂ and H ₂ O.

Performance Task:

Task Title: From Food to Fuel: Tracing the Path of Energy in Living Systems

Task Description: Learners will create an infographic or creative poster (digital or print) that explains how cells obtain nutrients from food (in animals) or from sunlight/CO₂ (in plants), and convert them into usable energy through cellular processes (photosynthesis and respiration).

The infographic/poster must show:

1. Inputs of photosynthesis and respiration (nutrients, gases, sunlight, etc.)
2. Processes inside the cell (chloroplast, mitochondria, glycolysis, Krebs cycle, etc.)
3. Outputs (ATP, oxygen, carbon dioxide, glucose, water)
4. Complementarity of photosynthesis and respiration



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5. Real-life Application: Explain one scenario (e.g., sports, farming, environment, or health) where these processes are essential for survival.

Output: A creative, visually engaging, and scientifically accurate infographic/poster.

Rubric for Performance Task (30 points total)

Criteria	Excellent (5)	Proficient (4)	Developing (3)	Beginning (1-2)	Weight
Scientific Accuracy	All processes correct, complete, well explained	Mostly correct, minor errors	Some inaccuracies; key parts missing	Major inaccuracies; little understanding	×3 (15 pts)
Organization & Clarity	Logical flow; well labeled diagrams	Clear, slight disorganization	Some confusion; incomplete labels	Very unclear; hard to follow	×2 (10 pts)
Creativity & Presentation	Highly creative, excellent design	Creative and neat	Limited creativity, plain design	Minimal effort	×1 (5 pts)

Total Score = 30 points